

## Sequencing proteins: Insulin

### A new research opportunity

Soon after Sanger was awarded his doctorate in 1944, Neuberger took up a position at the National Institute for Medical Research in London. Without Neuberger's expertise on tap, Sanger reluctantly realised he could no longer carry on working on lysine metabolism. The question, however, was what he should do instead.

Sanger's answer came from Albert Charles Chibnall, one of the examiners of his PhD, who had succeeded Hopkins on his retirement from the Sir William Dunn Institute of Biochemistry in 1943. Chibnall was an expert in plant biochemistry and specialised in the application of quantitative chemical analytical techniques to biology, a pursuit started by German organic chemists in the nineteenth century. His specific interest was the analysis of amino acids, the building blocks of proteins ([Garcia-Sancho, 2010](#)).

Having arrived in Cambridge, soon after Sanger was awarded his doctorate, Chibnall invited him to join the team he had brought with him from Imperial College, London. He wanted Sanger to work on a project directed towards unravelling the amino acid composition of proteins. This was a highly attractive proposition for Sanger. Firstly, it promised him his first paid research. So far he had been living on the inheritance from his mother. Secondly, it opened up an exciting new research avenue. Until then Sanger's focus had been on lysine, just one amino acid in proteins. Chibnall's proposal offered him an opportunity to extend his skills by working on the wider structure of proteins. Scientists had been keenly interested in proteins since the late nineteenth century, seeing them as molecules essential to virtually every biological process in the body. Working out the composition of proteins was thus seen as key to understanding how proteins functioned and the basis for their synthetic production([Garcia-Sancho, 2010](#)).

Up to then, 20 amino acids had been identified in proteins. Chibnall had himself identified five of them (aspartic acid, glutamic acid, arginine, histidine and lysine). The exact pattern and number of amino acids in each protein remained a mystery, however. As early as 1902 two German chemists, Franz Hofmeister and Emil Fischer had separately suggested a protein's amino acids were linked together by recurrent peptide bonds in a chain-like formation. This model, however, was not universally accepted. Some were sceptical, for example, about whether proteins were in fact pure chemical entities that had a molecular structure with a specific pattern. Indeed, many doubted whether peptide bonds alone could account for a protein's complex three-dimensional structure. An alternative explanation was that proteins were just a random mixture of material suspended in fluid ([Sanger, Dowding, 1996](#)).

By the time Sanger joined Chibnall's team, there was a growing consensus that proteins had a molecular structure. Nonetheless, scientists had little idea about which and how many of the naturally occurring amino acids were present in any given protein. Moreover, they did not know how they were arranged. Among their questions was whether they had a regular pattern or not. In a similar vein, they wanted to know whether amino acids were arranged in a ring-like formation or instead took the form of a unique linear sequence with two distinct ends. Considerable research was being undertaken on the physical properties of amino acids. Newly introduced ultracentrifuge equipment, for example, was being used to determine the molecular weight of amino acids and X-rays to study their structure. Some chemical techniques, such as colorimetry which measures the concentration of a chemical element with the aid of a colour reagent, were also being deployed. Most methods, however, were tedious and time-consuming and the results were unreliable ([Sanger, Dowding, 1996](#)).

## Insulin

One way Chibnall thought the structure of proteins and their amino acid composition could be understood was studying insulin, a small protein secreted by the pancreas that

helps regulate sugar levels in the blood. Insulin was attractive as a project for two reasons. Firstly, it was one of the few pure proteins then readily available. It was easily purchased in bottles from a local pharmacy. Secondly, insulin was of major medical interest, having been used for the treatment of diabetes since the 1920s, so pharmaceutical funding could be obtained for research on the protein. Thus Chibnall was soon able to attract funding from Eli Lilly and Company, the American pharmaceutical firm which helped develop insulin as a treatment for diabetes, and from Imperial Chemical Industries, a British chemicals company then in the process of building a pharmaceutical business ([Sanger, Dowding, 1996](#)).

Chibnall had already initiated his work on insulin while at Imperial College, and he now invited Sanger to join the project. Previous work by Chibnall's team indicated insulin had a much simpler composition than most proteins. Critically, it lacked two of the most commonly occurring amino acids in other proteins (tryptophan and methionine). Maurice Rees, Chibnall's chief assistant, had also discovered insulin contained a much higher content of one group of amino acids, known as alpha amino acids, than the team could account for. These amino acids appeared at just one terminal, labelled N, on the chain. Overall, the findings suggested insulin was made up of relatively short polypeptide chains. This meant the chains were potentially amenable to chemical analysis (Sanger, 1988).

## Identifying end amino acids

Sanger's first task was to identify the free amino acids at the N terminal of the insulin chain. Free amino acids are single molecules that are not bound by peptide bonds to other amino acids. A number of researchers had already devised some techniques to determine end-group amino acids in proteins. However, as yet, none of these techniques had produced any reliable results. Sanger first investigated a solubility product method developed by Max Bergmann and colleagues at the Rockefeller Institute, New York. He soon rejected it, however, because it necessitated very accurate

weighing of many small samples which was extremely laborious given the weighing equipment of the time (**Sanger, Dowding, 1996**).

He settled instead on partition chromatography. This technique had been developed in the early 1940s by Richard Synge and Archer Martin, two former research students at the Dunn Institute of Biochemistry, who had moved to Leeds to work for the Wool Industries Research Institute. They had devised a partition chromatography method to determine the composition of amino acids in wool. This research project was part of a wider effort to revive the wool market which was then undergoing a slump due to the emergence of synthetic fibres (de Chadarevian, 1996; Sanger, Dowding, 1996).

Publishing their method in 1941, Synge and Martin's technique involved packing a tube tipped vertically with ground up silica gel, then wetting the gel with water and pipetting in an amino acid solution at the top. Chloroform was then inserted to wash the amino acid solution through. A methyl red dye was also added. This dye formed bright red bands against an orange background which helped show up the separated amino acids.

Overall, partition chromatography was considered far superior to all previous fraction methods. It was also being used by George Roland Tristram in Chibnall's group to work out the amino acid composition of insulin (Ettre, 1991; Ettre, 1994; de Chadarevian, 1996; Shetty, 1993).

Having worked out what separation technique to use, Sanger needed to find a suitable label for the amino acids. So far only one tag mentioned in the literature had successfully identified an amino acid at the N terminal of a protein chain. The first reagent Sanger tested was methane sulfonyl chloride. This reagent, however, proved disappointing. After this Sanger investigated dinitrochlorobenzene, an organic compound. Encouragingly this stained the amino acids bright yellow making them stand out in a clear band when subjected to partition chromatography. The reagent, however, had a major disadvantage. It only bound with amino acids when heated. This was a problem because of the tendency of heat to destabilise compounds, which could undermine the ability of any resulting derivatives to withstand acid hydrolysis, the

process required for breaking apart the peptide bonds between the amino acids (de Chadarevian, 1996; Sanger, 1992).

After Sanger had failed with a number of different reagents, which were mostly chlorine based compounds, Chibnall suggested he try a fluoro compound. This was a radical suggestion. Most organic chemists were wary of working with fluoro reagents because of their high toxicity. Indeed, many scientists had been burned and even killed using them. Chibnall's proposition, however, had three attractions. Firstly, fluoro compounds stain things yellow. Secondly, Chibnall suspected they would bind to amino acids without the need for heat. Lastly, their availability was increasing because a number of scientists were synthesising fluoro compounds as part of military research into chemical weapons (de Chadarevian, 1996; de Chadarevian, 1999).

Chibnall soon sourced a reagent for Sanger to try. This was fluorodinitrobenzene (FDNB), a compound first prepared in 1904. It was being synthesised by the organic chemist, Bernhard Charles Saunders, in Cambridge University's Department of Chemistry, as part of research for the Ministry of Supply into the physiological effects of poisonous gases. Encouragingly Sanger found that FDNB reacted with amino acids at room temperature. The resulting dinitrophenyl-amino-acids also proved more stable than the peptide bonds when subjected to heat hydrolysis and appeared bright yellow when put into a solution. This was important for the next stage – partition chromatography (de Chadarevian, 1996).

The first experiments Sanger conducted were with an amino acid called glycine, but the results were not decisive. Surprisingly, when subjected to partition chromatography the FDNB glycine derivative showed up in two bands instead of just one. Sanger spent a considerable amount of time working out why this happened. It was only later, after he had tagged other amino acids with FDNB, which showed up in just one strong band, that he worked out that glycine was an exception (Sanger, 1988).

By 1945, Sanger had developed a three stage method for identifying, quantitatively measuring and characterising the terminal amino acids in insulin. This involved treating

the protein with FDNB, subjecting it to acid hydrolysis and then separating out the coloured compounds with chromatography. His technique marked a major improvement on early efforts to determine end group amino acids. Importantly, it made it possible to estimate the number and length of peptide chains in proteins which was fundamental to determining a protein's structure. Overall, Sanger had identified two end-group amino acids in insulin: glycine and phenylalanine. This suggested insulin had four open peptide chains. Two ended with the amino acid called phenylalanine and the other two ended with the glycine amino acid. From this it appeared that insulin possessed only two types of chains, and not 18 as Chibnall had originally hypothesised. He was able to establish that the two chains were linked together by cystine, another amino acid (Sanger, 1945).

#### A shift in focus: All amino acids in insulin

Sanger had successfully separated the two chains of insulin by using performic acid by 1947 (Sanger, 1947). This made it possible for him to start analysing the full range of amino acids in each chain. Undertaking a fuller structural analysis marked a radical departure from the original task set by Chibnall to identify just the handful of end-group amino acids, and the project played to his strengths as a pure bench scientist. He was much more interested in the design and application of different methods than theorising and carrying out the abstract projects which were the priority of other members in the department ([Garcia-Sancho, 2010](#)).

Analysing amino acids further along the two insulin chains posed a major challenge, however. Most of the available separation techniques could not fractionate a product as large as the two insulin chains.

In 1947, whilst he was debating how to proceed, Sanger was invited to work in the laboratory of the Swedish biochemist Arne Tiselius. Based at the Fysikalisk-kemiska Institutionen in Uppsala, Tiselius and his team were at the forefront of advancing an electrical separation method for analysing proteins known as electrophoresis. This

technique rested on a late nineteenth century scientific discovery that when molecules were placed in an electric field they moved in different directions and at varying speeds according to their electric charge, physical motility and size. Tiselius had taken this technique a stage further based on his observation in the late 1920s that an electrical current caused purified proteins to migrate as homogenic bands. By 1937 he had developed an electrophoresis apparatus for separating proteins and enzymes. Being about five metres in length, this instrument took up the space of a medium-sized laboratory. This work made Tiselius's laboratory a magnet for many visitors, including some leading American immunologists ([Wollheim, 2014](#)).

Sanger eagerly accepted the invitation to spend time with Tiselius and his team. The proposition had two attractions. Firstly, it made it possible to spend time in Sweden, which, unlike Britain, was not in the grip of post-war austerity, shortages and rations. Secondly, and more importantly, it offered him an opportunity to learn the techniques being advanced by Tiselius([Sanger, 1988](#)).

What Sanger was to value most from his visit to the Swedish laboratory was his encounter with Synge then visiting from Leeds. Critically, Synge taught him zone ionophoresis on starch, another fractionation technique, which used an electrical current to separate a mixture's components based on their ion and polar charge ([Sanger, 1988](#)).

Once back in Cambridge, Sanger serendipitously found that he could break up the protein into mini-chains, made up of four to five amino acids, from near the N termini of the two insulin fragments. This he achieved by diluting the strength of the acid used for hydrolysis and reducing the exposure time to the acid. The advantage of the mini-chains was that they were amenable to separation by the newly developed technique called paper chromatography, an off-shoot of partition chromatography which had been published in 1944 by Archer Martin and two other colleagues based at the Wool Industries Research Institute in Leeds. Paper chromatography represented a major

improvement on previous chromatographic methods. The procedure entailed putting a drop of an amino acid solution on the edge of a strip of filter paper wetted with water and then dipping that paper into a solvent. Once absorbed the solvent spread across the paper in two different directions carrying with it the mixture's components. After this the paper was dried and sprayed with ninhydrin, a colouring reagent that reacts with proteins. With the components moving at different speeds on the paper it became possible to see them as distinct and physically separate spots. Critically paper chromatography could be performed with just basic equipment and several samples could be analysed simultaneously (Consden et al, 1944).

## Sequencing insulin's two chains

By 1949 Sanger was ready to begin analysing the composition of the two chains of insulin. The first chain, labelled A, had glycine at its terminal, and the second, labelled B, had phenylalanyl at its end. Initially Sanger aimed to investigate the two chains simultaneously. It soon became clear, however, that fraction A, although the shorter of the two, would be more difficult to analyse. Sanger therefore decided to focus his efforts first on fragment B ([Sanger, 1958](#)).

Much of the work on chain B was undertaken by Hans Tuppy, a biochemist researcher who joined Sanger from Austria in 1949 or 1950. At first their progress was hampered by the fact that acid hydrolysis did not produce sufficiently long chain fragments. After much deliberation, they decided to switch from acid to enzyme hydrolysis. They hesitated to do this initially because other researchers in the department, who had expertise in enzyme kinetics, had warned them that enzymes could synthesize peptide bonds as well as break them apart and so had the potential to rearrange the amino acids in the fragments. Sanger and Tuppy, however, found that hydrolysis with a proteolytic enzyme produced sufficiently large fragments without affecting their amino acid arrangement ([Sanger, 1988](#)).

Within a year Tuppy had identified and determined the sequence of all 30 amino acids in chain B (Sanger, Tuppy 1951). Sanger attributed this achievement to Tuppy's very hard work. As he recalled: 'Much of the work required the use of paper chromatography; the chromatography room was at the other end of the basement of the corridor, and a common sight at that time was Tuppy walking at full speed along that corridor bearing chromatograms. He never ran, but to keep up with him anyone else had to run' (Sanger, 1988).

Following Tuppy's departure when his fellowship ended, Sanger set to work on chain A, assisted by Ted O P Thompson, his Australian doctoral student. Determining the structure of this chain proved much more difficult than for chain B. Identifying the amino acids in chain B had been helped by the fact that several of its amino acids occurred just once in the molecule, usually near the end of the chain. By contrast chain A had fewer unique amino acids. This made it more difficult to determine their pattern. By 1953, however, Sanger and Thompson had succeeded in sequencing all 21 amino acids in chain A (Sanger, Thompson 1953).

After this success, Sanger began working out the composition of the amino acids in the disulfide bridges, the chemical bonds that linked the two chains together, with the help of A P Ryle, Leslie F Smith and Ruth Kitai. Work on this proved especially difficult. As Sanger recalled, it 'involved many man-hours, and more frustrations' than all the work he had done so far on the two previous chains. Among their difficulties, they found that the acid hydrolysis tended to rearrange the disulfide bridges. After some trial and error with acid hydrolysis, however, they overcame the problem (Ryle et al, 1955). [Click here for some of Sanger notes on insulin's disulphide bridges.](#)

By 1955 Sanger and his team had sequenced all 51 amino acids in the two chains of insulin and worked out the position and composition of the three disulfide bridges which joined them (Sanger et al, 1955). Getting to this point had been a painstaking and time-consuming process. It involved breaking insulin up into small fragments and then

reconstituting its chains by identifying where its amino acids overlapped. Sanger described the process like piecing together a jig-saw. His technique would later be called the degradation or DNP method.

## The novelty of Sanger's sequencing technique

Many of the elements of Sanger's degradation technique were not new. For example, the means he used to cleave and separate the amino acids and then quantify them paralleled some of the analytical chemistry methods deployed by Chibnall and his team. Similarly, Sanger's use of overlaps in the fragments to reconstitute the insulin molecule mirrored the way synthetic chemists worked. What was novel about his method, however, was his introduction of partial rather than complete hydrolysis as well as his use of DFNB for tagging the amino acids and paper chromatography. All of these were important innovations because they provided a way of breaking down a complex protein into manageable fragments, and this facilitated the precise identification of its amino acids and their sequence ([Garcia-Sancho, 2010](#)).

Sanger's gradual shift away from only studying amino acids on the ends of the insulin chain to sequencing the whole structure of the protein had led from his gradual awakening to the power of sequencing. The idea of sequencing was also not totally new. Sequencing had been a major goal of protein chemists since the early twentieth century. More recently, in 1943 Martin and his colleagues in Leeds suggested that sequencing proteins could be achieved through the use of partial hydrolysis and paper chromatography, and three years later had demonstrated its power by sequencing gramicidin S, an antibiotic (Consden et al, 1947). Comprised of just five amino acids, gramicidin S was, however, less laborious to sequence than insulin, which was much bigger. Sanger's work on insulin also differed from that undertaken by the Leeds researchers because he began to see sequencing as a research goal in and of itself. By contrast, the sequencing effort of the Leeds researchers was designed to test separation techniques (de Chadarevian, 1966; Garcia-Sancho, 2010).

In 1958 Sanger was awarded the Nobel Prize for his achievements in sequencing insulin. Importantly, he had demonstrated conclusively for the first time that proteins were real chemicals with a defined sequence. His work also put an end to previous theories which suggested that amino acids in proteins did not have any kind of regular arrangement (Garcia-Sancho, 2010). Inspired by Sanger's results, many other scientists began to sequence amino acids in other proteins.

### Improving the degradation sequencing method

Overall the process of completing the sequencing of insulin had been time-consuming and painstaking. Well before Sanger had received the Nobel Prize it was clear to both him and others that further separation fraction techniques would be needed to handle larger more complex chains than insulin, which had relatively short chains.

In 1954 Sanger developed a new way to conduct his degradation method based on contact with Chris Anfinsen, an American biochemist then spending a sabbatical year in the Department of Biochemistry. Importantly, Anfinsen introduced him to the technique of attaching isotopes, radioactive substances, to proteins to act as labels. The technique made it possible to tag a sample with an isotope which could be then placed on a surface ready for separation by an electrical current. Following separation, the sample was covered with a photographic emulsion for a short time. The emulsion, usually a gel containing silver halide, was then developed, fixed, and washed in the same way as a photograph. In this process the photographic film darkened in places where silver halide grains were present. The pattern formed by the grains on the film depended on how the radioactively labelled components in the sample separated and the type of radiation and nature of the photographic film. The end product, a picture, was called a radiogram or autoradiograph (Garcia-Sancho, 2010).

Such methods had been commonly deployed by physiologists and scientists involved in nuclear medicine from the 1930s, but before meeting Anfinsen Sanger assumed that

only physicists could handle isotopes, and that autoradiography was beyond him. He also feared that the incorporation of isotopes into protein fragments would rearrange their amino acids. His reticence also reflected his intense dislike of physics, a hangover from his undergraduate days. Sanger had also been discouraged by the fact that early on in his insulin work he had attempted, with little success, to use iodine, a radioactive substance, to tag the amino acid tyrosine ([Sanger, 1988](#); [Garcia-Sancho, 2010](#)).

On watching Anfinsen, Sanger soon realised that if used in conjunction with paper chromatography, radioactive isotopes and autoradiographic techniques could provide a powerful and simple means to reduce some of the tedious steps involved in the analysis of amino acids, an important chemical step in sequence determination. Up to this time the only way this could be achieved was by applying a series of reagents to the chromatogram, to render the amino acids visible after their separation, and then carrying out the time-consuming and laborious process of identifying and quantifying them ([Sanger, 1988](#); [Sanger, 1992](#); [Garcia-Sancho, 2010](#)).

By the late 1950s Sanger had begun to experiment with isotopes acquired from a radiochemical centre specialising in the manufacture of radioactive tracers that was attached to the British Atomic Energy Authority, a government body supervising the peaceful development of nuclear technology. Based on this work Sanger soon began producing autoradiographs which showed the amino acids forming distinct patterns. These patterns he called 'protein fingerprints'. The advantage of the technique was that by measuring the position of the spots in the picture Sanger could now deduce the sequence of peptides without having first to undertake a full amino acid analysis. Nonetheless, he was not wholly satisfied with the technique ([de Chadarevian, 1996](#); [Garcia-Sancho, 2010](#)).

Sanger's degradation technique was soon superseded by one developed by Pehr Edman, a Swedish biochemist based at the University of Lund. As was the case in Sanger's method, Edman's technique involved first cleaving the protein, but this was

done with a chemical reagent instead of acids and enzymes. The reagent he used was phenylisothiocyanate. One of the advantages of Edman's technique was that it cut down on the number of steps in the procedure. Importantly, it allowed sequences to be determined unit by unit in their correct order. By 1958 Stanford Moore and William Stein, two American biochemists based at the Rockefeller Institute for Medical Research, had developed an automatic analyser based on Edman's method.

Following this a number of proteins were successfully sequenced during the early 1960s. This included 1) pancreatic enzyme ribonuclease; 2) trypsin and chymotrypsin, proteolytic enzymes which help break down proteins in food and in mammals' intestines; 3) two chains of haemoglobin, a protein that carries oxygen in the blood; and 4) myoglobin, the carrier of oxygen in muscles ([de Chadarevian, 1999](#)).

## The trail to sequence alignment methods

By the time Sanger had completed his analysis of the structure of insulin he was totally gripped by the research possibilities sequencing offered. As he put it, the 'sequencing bug had really taken hold of me' ([Sanger, 1988](#)). Critically he believed it could help unravel the function of proteins. It soon became, clear, however, that in the case of insulin knowing a complete sequence revealed little about how a protein worked. One way he thought he could resolve the puzzle was to identify the 'active centre' of insulin. To this end he began to explore the ways he could label the active centre of insulin and then determine the sequence around it. Another approach he took was comparing the sequences of insulin taken from five different species. Until then he had only worked on insulin taken from cattle. Once again, he worked with collaborators ([Sanger, 1988](#)). The work was published in Brown, H, Sanger, F, Kitai, R (1955) 'The structure of pig and sheep insulins', *Biochemistry Journal*, 60/4: 556-65.

While such efforts uncovered some amino acids that were not essential for activity, they did not get Sanger any closer to understanding insulin's mechanism of action.

Nonetheless, the work revealed that different proteins had very similar but not identical

amino acid compositions. The differences were all located in one small segment on the glycyl chain. This was the first evidence that proteins shared significant similarities which indicated their evolutionary relationships. The discovery laid the foundation for the development of sequence alignment methods, a procedure commonly used by bioinformatic scientists today for comparing and finding similar sequences of amino acids or DNA base pairs. Aided by computers since the 1970s, this technique helps in the classification of genes and proteins and also helps to determine their biological function, detect point mutations and construct evolutionary trees (Crick, 1958; de Chadarevian, 1999; Lagnado, 2014).

### Away from the laboratory

When not busy in the laboratory, Sanger enjoyed taking time out to play squash and cricket. He also enjoyed a rich family life. By the time he received his Nobel Prize his two sons, Peter and Robin, were aged 12 and 15. Sanger fondly attributed his scientific success to his wife Joan. He believed her maintenance of their peaceful and happy home gave him the freedom to pursue his scientific interests.

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